

Computer Networks Spring 2026

Assignment#1 (6A & 6B)

Due Date: 3-February-2026

Submission Mode & Time: Handwritten solutions to be submitted during the lecture.

Please note the following:

1. No exceptions to the above date and time will be allowed. Inability to submit the assignment by the required time will result in zero marks.
2. To ensure self-completion of assignments and discourage plagiarism, the instructor or the relevant TA may randomly contact you and ask for an explanation of your answers. Where plagiarism and/or cheating is evident, you will be referred to the departmental disciplinary committee. In extreme cases of plagiarism an F may be awarded immediately with further referral to the university disciplinary committee.
3. All solutions must be **hand-written**.
4. **Assignment Solution Submission:** In case of **in person / physical lectures at the campus**, hard copy of the hand-written assignment's solutions will be submitted by **hand** by each student to the Instructor / TA directly during the lecture on the due date.

PART-1

Use the following text for completion of this part of the assignment:

Computer Networking - A Top-Down Approach 8th Edition by Kurose & Ross.

Solve the following problems from the back of **Chapter 1**. Every Question has equal marks i.e.

Review Questions: (4+4 = 8 marks) [CLO 1]

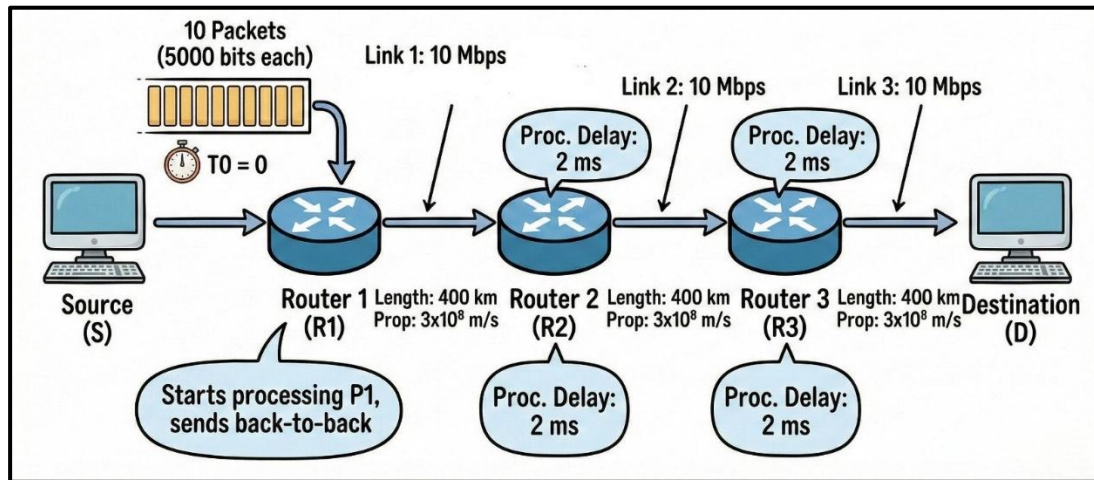
R12, R19

Problems: (4+4+4 = 12 marks) [CLO 1]

P3, P6, P10

PART - 2

Question: 01 [10+10+3 =23 Marks] [CLO 1]



Suppose that 10 identical packets each comprising 5000-bits are sent back-to-back from source node to destination node with three routers between them making the path as $S \rightarrow R1 \rightarrow R2 \rightarrow R3 \rightarrow D$. Thus, there are 4 links in total. Now, suppose that each link has a transmission rate of 10 Mbps, propagation speed for each transmission link is 3×10^8 m/s while length of each links is 400 km. Consider that each router incurs a per packet processing delay of 2 ms.

Moreover, assume that all packets are well received and buffered by router R1, and the current time is 0 (called T_0). Router R1 starts processing of the first packet at T_0 and sends packet one after another. Router R2 incurs the given processing delay on each packet and forwards the same toward router R3. For this scenario, answer the following questions assuming that there is neither any packet loss nor any queuing delays:

(1a): At what time will each packet be leaving router R1 (T_1), at what time will each packet be received by router R2 (T_2), at what time will each packet be leaving router R2 (T_3) and at what time will each packet be received by router R3 (T_4). For the given scenario, write the first 5 entries (row 1 to 5) as per the following sample table format for the first 5 packets (packets 1 to 5). Remember that the initial time (T_0) is 0.

Packet #	T_1	T_2	T_3	T_4

(1b): Now suppose that the transmission rate of sender dropped to 75% of its original transmission rate after forwarding 5 packets. Considering this condition, write the next 5 entries (row 6 to 10) for the remaining 5 packets (packets 6 to 10) considering the new transmission rate:

(1c): Compare the total delays for a single packet in both cases. What is the percentage increase in delay when the transmission rate is reduced to 75%?

Note: A packet is considered as leaving a node after necessary processing and transmission delays. Also, please assume that the processing of the next packet starts only after the previous one is fully transmitted.

Question: 02 [6 Marks] [CLO1]

Explain the difference between link bandwidth and end-to-end throughput. Create a scenario (draw a simple diagram described in text) where a high-bandwidth core network still results in low throughput for the user.

Question: 03 [3+2+1=6 Marks] [CLO 1]

Consider a communication link shared by N users. The link capacity is 2 Mbps. Each user, when active, requires 200 kbps.

- **Case A (Circuit Switching):** How many users (N_{cs}) can be supported simultaneously using strict TDM circuit switching?
- **Case B (Packet Switching):** Suppose users are bursty: they are active only 10% of the time ($p=0.1$).
 1. If we have 30 users, what is the probability that exactly 15 users are transmitting simultaneously? (Set up the binomial formula; you do not need to solve the final decimal).
 2. Explain why packet switching allows us to support more than the N_{cs} calculated in Case A with a very low probability of congestion.

Question: 04 [2+2+1=5 Marks] [CLO 1]

You are downloading a file from a server in the US to a laptop in Pakistan.

- **Server Access Link:** 1 Gbps
 - **Trans-Atlantic Backbone Link:** 100 Gbps (shared by 50,000 active flows)
 - **Pakistan ISP Core Link:** 10 Gbps (shared by 2,000 active flows)
 - **Your Home Wi-Fi Link:** 5 Mbps
- a. Estimate the available bandwidth per flow on the Backbone link and the ISP Core link (assuming fair sharing).
 - b. Identify the bottleneck link in this path.
 - c. What is the maximum theoretical throughput you can achieve for this download?